



Review

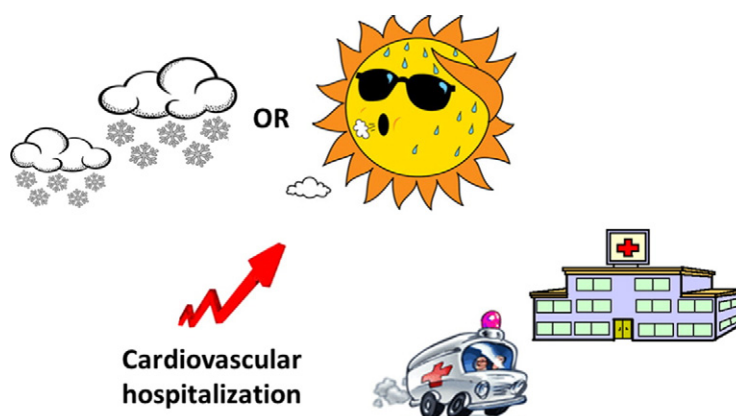
Ambient temperature and risk of cardiovascular hospitalization: An updated systematic review and meta-analysis

Dung Phung^{a,*}, Phong K. Thai^{b,**}, Yuming Guo^c, Lidia Morawska^b, Shannon Rutherford^a, Cordia Chu^a^a Centre for Environment and Population Health, Griffith University, Australia^b International Laboratory for Air Quality & Health, Queensland University of Technology, Australia^c Division of Epidemiology and Biostatistics, School of Public Health, University of Queensland, Australia

HIGHLIGHTS

- There are significant short-term effects of cold, heatwave to CDV hospitalization.
- Similar association was observed for diurnal temperature variation.
- There is an inconsistent effect of heat exposure on CDV hospitalizations.
- Future studies need to focus on specific geographical and climate areas.

GRAPHICAL ABSTRACT



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ABSTRACT

The association between temperatures and risk of cardiovascular mortality has been recognized but the association drawn from previous meta-analysis was weak due to the lack of sufficient studies. This paper presented a review with updated reports in the literature about the risk of cardiovascular hospitalization in relation to different temperature exposures and examined the dose–response relationship of temperature–cardiovascular hospitalization by change in units of temperature, latitudes, and lag days. The pooled effect sizes were calculated for cold, heat, heatwave, and diurnal variation using random-effects meta-analysis, and the dose–response relationship of temperature–cardiovascular admission was modelled using random-effect meta-regression. The Cochrane Q-test and index of heterogeneity (I^2) were used to evaluate heterogeneity, and Egger's test was used to evaluate publication bias. Sixty-four studies were included in meta-analysis. The pooled results suggest that for a change in temperature condition, the risk of cardiovascular hospitalization increased 2.8% (RR, 1.028; 95% CI, 1.021–1.035) for cold exposure, 2.2% (RR, 1.022; 95% CI, 1.006–1.039) for heatwave exposure, and 0.7% (RR, 1.007; 95% CI, 1.002–1.012) for an increase in diurnal temperature. However no association was observed for heat exposure. The significant dose–response relationship of temperature – cardiovascular admission was found with cold exposure and diurnal temperature. Increase in one-day lag caused a marginal reduction in risk of cardiovascular hospitalizations for cold exposure and diurnal variation, and increase in latitude was associated

* Correspondence to: D. Phung, Griffith University, Australia.

** Correspondence to: P. K. Thai, Queensland University of Technology, Australia.

E-mail addresses: d.phung@griffith.edu.au (D. Phung), phong.thai@qut.edu.au (P.K. Thai).

with a decrease in risk of cardiovascular hospitalizations for diurnal temperature only. There is a significant short-term effect of cold exposure, heatwave and diurnal variation on cardiovascular hospitalizations. Further research is needed to understand the temperature-cardiovascular relationship for different climate areas.

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1. Introduction

There is increasing evidence that on-going global climate change has generated more frequent and intense extreme weather events such as heatwaves and cold spells (IPCC, 2013). Such extreme temperatures have been associated with significant health impacts. For example, excessive hospitalizations due to a range of morbidity such as heat stroke, cardiovascular and respiratory diseases have been recorded during heatwave episodes (Empana et al., 2009; Knowlton et al., 2009; Ma et al., 2011; Semenza et al., 1999; Turner et al., 2013). The serious impacts of heatwaves on human health have led to the first guidance for a Heat-Health Warning system jointly developed by the World Meteorological Organization and World Health Organization (McGregor et al., 2015). On the other hand, episodes of extreme cold are major health threat in high-latitude countries (Huynen et al., 2001; Kysely et al., 2009; Pattenden et al., 2003; Shaposhnikov et al., 2014). Previous studies have indicated that one of the predominant causes of hospitalizations associated with extreme temperature is cardiovascular diseases (CVD) (Bayentin et al., 2010; Ebi et al., 2004; Oshige et al., 2006; Schwartz et al., 2004; Turner et al., 2012a) which can cause a large burden for the health system. Nevertheless, most environmental-related research to date has concentrated on the relationship between temperature and cardiovascular mortality (Baccini et al., 2008; Basu, 2009; Gasparrini et al., 2015; Guo et al., 2014; Huang et al., 2012; McMichael et al., 2008; Son et al., 2015; Yang et al., 2015).

Some systematic reviews have been recently conducted to evaluate the relationship between ambient temperature and morbidity, in which cardiovascular hospitalization has been involved as one of the morbidity causes (Astrom et al., 2011; Bhaskaran et al., 2009; Turner et al., 2012b; Ye et al., 2012). However, these studies have limitations which might result in inconclusive findings about temperature-CVD hospitalization relationship. Turner et al. (2012b) conducted a meta-analysis to evaluate the association between temperature and cardiorespiratory morbidity using a small number of studies, and the influences of cold exposure, heat waves, and variation of diurnal temperature on CVD hospitalizations were not analysed in this study. Bhaskaran et al. (2009) conducted a systematic review of the influence of temperature on the incidence of myocardial infarction with findings about the effects of both hot and cold weather on the risk of Myocardial infarction. Ye et al. (2012) systematically reviewed epidemiological evidence on temperature-

morbidity relationship and indicated that there was a significant short-term effect of ambient temperature on total and cause-specific morbidities, including cardiovascular diseases. These two studies nevertheless did not quantify the pooled effect sizes of temperature-CVD hospitalization relationship. Another review by Astrom et al. (2011) found that the evidence of temperature-cardiovascular morbidity relationship among elderly people was inconclusive, and this study also did not quantify the pooled effects of temperature-CVD hospitalizations.

It is thus important to address this gap of information with newly published works in the literature. This study aims to conduct extensive search with the objectives to: (i) investigate the risk of cardiovascular hospitalization in relation to different temperature exposures (cold, heat, heatwaves, and diurnal temperature), and (ii) examine the dose-response of temperature-CVD relationship by change in units of temperature, latitudes, and lag days.

2. Method

This meta-analysis was conducted according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement (Moher et al., 2009) and the recommendations of the meta-analysis of Observational Studies in Epidemiology group (MOOSE) (Stroup et al., 2000).

2.1. Study selection

The PubMed electronic database was used to search published studies examining the relationship between ambient temperature and cardiovascular morbidity. In order to avoid missing relevant studies, the keywords and Medical Sub-Heading (MeSH) terms in the initial search were set in two categories with 'or' operand used within a group and later 'and' or 'not' operands used between groups. The two categories comprised: (i) exposure variables, namely: climate change, temperature, heat wave, heatwave, cold, cold spells; (ii) outcome variables, namely: health effects, morbidity, hospitalization, emergency department visit, death, mortality. Our search was limited to 'English' and 'Human' studies. Then a further screening was conducted through the abstract review to identify the studies concerning the temperature-cardiovascular relationship (Fig. 1).

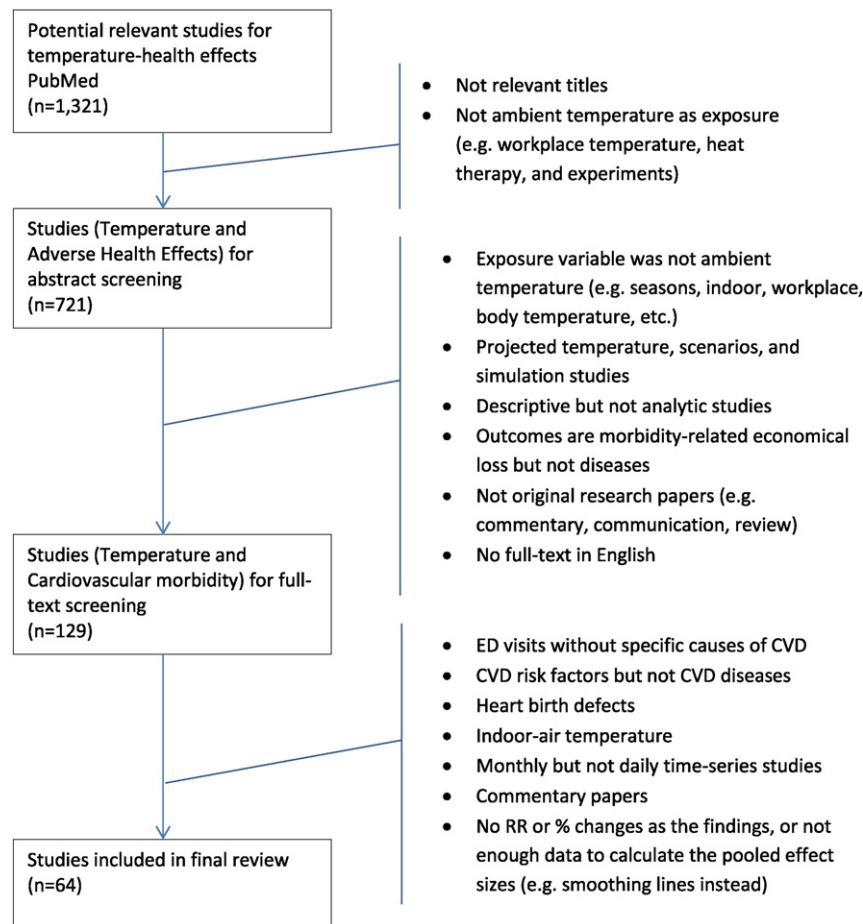


Fig. 1. Flow diagram for inclusion and exclusion of studies.

Three criteria for eligibility were used. First, published analytic studies needed to be based on one of the following calculations: regression coefficient, percentage change, relative risk (RR), and odds ratio (OR) on the relationship between ambient temperature and cardiovascular morbidity. Second, the study designs were required to be appropriate for examining the short-term effects of changes in temperature on cardiovascular morbidity (daily timescale), so each study had to contain an outcome measure related to hospitalizations for either all-cause or specific-cause cardiovascular diseases. Third, exposure to temperature was based on at least one of the following conditions:

- *Cold exposure*: number of degrees below the defined threshold or average value, or comparison between extreme cold condition and the reference value (e.g. 5th vs. 90th percentiles);
- *Heat exposure*: number of degrees above the defined threshold or average value but NOT during the cold season (e.g. winter), or comparison between extreme hot condition and the reference value (e.g. 99th vs. 75th percentiles);
- *Cold spells*: ≥ 2 days of extreme low temperature (e.g. 5th percentiles);
- *Heat wave*: ≥ 2 days of extreme high temperature (e.g. 95th percentiles);
- *Variation in diurnal temperature*: number of degrees between maximum and minimum temperature during the day.

The evidence from previous studies (Barnett et al., 2010; Basu et al., 2008; Hajat and Kosatky, 2010) demonstrated that the influence of temperature on mortality does not vary substantially with the temperature

measures used. Therefore, all temperature measures were allowed in this study.

Criteria for exclusion included: (i) studies providing qualitative evaluation; (ii) studies providing only non-linear curves of temperature–outcome relationship because we cannot extract the effect sizes from these studies; (iii) studies using indoor or workplace temperature as the exposure variable; (iv) reviewing or commentary papers; (v) published papers with only English abstracts but without full texts in English.

The screening of potentially eligible studies was carried out using the tier approach with sequential steps (Fig. 1). First, the potential relevant studies for temperature–health effect relationship were obtained from a search of a wider study in temperature–health effects. Second, all obtained papers were title reviewed to check if their titles were relevant to the inclusion criteria or met exclusion criteria. Third, abstracts of selected papers were reviewed against the inclusion criteria. Fourth, the full texts of the papers selected from step 3 were reviewed against the inclusion criteria, and additional papers were identified and reviewed using manual searching in the references of full-text candidate papers.

2.2. Quality assessment

The criteria for evaluating the quality of the selected studies was modified from the criteria recommended by the BioMed Central for study assessment (Berman and Parker, 2002) (Table S2). The criteria used for evaluating quality of the studies comprise: (i) sources of the information including the time of data collection; (ii) study design, including clarity, appropriateness of, clear measurement of exposure variables, statistical packages, controls for confounding factors, and

the clear definition of health outcomes; (iii) valid interpretation of results; and (iv) discussion. We developed a score scheme corresponding to the assessment criteria, and each study was evaluated using this score system. The quality score could range from 1 to 27. A cut-off point of 13 was used for a study to be included in this meta-analysis. As indicated in Table S2, a study that has score of more than 50% (13) was published in a peer-review journal, has well-described study design (clear research questions and descriptions in temperature exposures), and its results properly answered the research questions with presented measurements (RR, OR).

2.3. Data extraction

The data which were extracted from each of the included studies comprise: first author last name and publication year, country and location in which the study was performed, study design and time-span, temperature measures as exposure variables, threshold or reference temperature, day lags of the effect, outcome (% change in risk/RR/OR and 95% CI, cardiovascular diseases diagnosed), and controlled variables.

2.4. Data synthesis and analysis

Statistical analyses involved a two-step approach: (i) computing pooled effect sizes for each type of temperature exposure using a random-effect meta-analysis approach; (ii) modelling the dose-response pattern of temperature-cardiovascular relationship using a random-effect meta-regression analysis.

In step 1, effect estimates were converted to a relative risk (RR) reflecting a change in hospitalizations due to a change in the condition of temperature exposures (cold, heat, heat wave, and variation in diurnal temperature). Standard errors for the relative risk were derived from associated confidence intervals. Because studies were conducted in different populations, resulting in considerable heterogeneity of findings, the random effects methodology (Borenstein et al., 2010) was applied to calculate for within-study and between-study variation and generate pooled effect sizes. The pooled effect sizes were calculated separately for each type of temperature exposure: cold, heat, cold spell, heatwave, and variation in diurnal temperature. Due to the high variability of involved factors such as lags, latitude, and population, the use of effect size from each included study may not fully represent the relative magnitude of underlying risk. We applied the Empirical Bayesian (EB) approach (Clayton and Kaldor, 1987) to adjust the effect sizes of the included studies by incorporating data from other spatial settings using the following equations:

$$\theta_j = \gamma_j \bar{y}_j + (1 - \gamma_j) \bar{y} \quad (1)$$

$$\gamma_j = \frac{\tau^2}{\tau^2 + \sigma_j^2} \quad (2)$$

where, θ_j is the effect size of each included study j ; γ_j is the shrinkage factor; τ^2 is the variance between studies; and σ_j^2 is the variance within each included study j . Then the pooled effect sizes were computed and compared with those obtained from the normal meta-analysis.

Heterogeneity between studies was quantified using Cochran's Q -statistics by summing the squared deviations of each study's estimate from the overall meta-analytic estimate, and weighting each study's contribution in the same manner as in the meta-analysis. $P < 0.05$ was deemed significant (Cochran, 1954). However, the Q -statistics is susceptible to the number of studies included in meta-analysis. Therefore, we used the coefficient of inconsistency (I^2) recommended by Higgins et al. (2003), which can provide a measure of the degree of inconsistency in the studies' findings by describing the percentage of total variation across studies that is due to heterogeneity (Higgins et al., 2003). The

heterogeneity degree ranged from 0 to 100%. The results were categorized as low ($<25\%$), moderate (25–75%) and high ($\geq 75\%$) heterogeneity. To investigate the possibility of publication bias, we used the approaches recommended by Egger et al. (1997); Sterne and Egger (2001) and Sterne et al. (2001). We first inspected the Begg's funnel plot with pseudo 95% confidence limits, which plots a measure of study precision (standard error as a function of effect size). Visual inspection of a funnel plot provides an indication of publication bias when larger and smaller studies are asymmetrically distributed across the combined effect size (Rothstein et al., 2005). The existing publication bias was further tested using Egger's test which evaluates the bias captured by the funnel plot by regression of the standard normal deviation on precision, defined as the inverse of the standard error.

In step 2, for the meta-regression analysis, we modelled the change in effect sizes (risk of cardiovascular hospitalizations) by the change in temperature degrees, lag days, and absolute latitude of the population using Eq. (1):

$$\theta_{ij} = \beta_0 T_{ij} + \beta_1 \text{Lag}_{ij} + \beta_2 \text{Lat}_{ij} + \varepsilon \quad (3)$$

where, θ_{ij} is the effect size of each study; T_{ij} is the change in temperature measures ($^{\circ}\text{C}$); Lag_{ij} is the lag days; Lat_{ij} is the latitude; and ε is the residue. If the study compared the effect sizes between two specific temperature conditions (e.g. 0°C vs. 15°C , or 5th vs. 75th), the difference value between the two compared points was used as the continuous change in temperature measures. For the cumulative lags (e.g. 0–1, 2–15, 16–27), the midpoint (arithmetic mean) was assigned as the continuous change in lag days. A previous study (Turner et al., 2012b) found that a polynomial model for the lag effect did not perform better, so a linear term was used. Separate analyses were performed on studies related to cold and heat exposure using random-effect regression with empirical Bayesian technique.

Some sensitivity analyses were also performed. First, the pooled effect size was calculated after excluding the lower quality studies which have an assessment score of less than 75% of the total maximum score. Second, the pooled effect sizes were computed for the same-day effect only (lag 0), since the temperatures have been observed to have immediate and short-term effects on hospitalizations. Finally, the pooled effect size of hospitalizations only was estimated after excluding the hospital consultations and CVD biomarkers as the health outcome.

3. Result

3.1. Included and excluded studies

A total of 1321 papers in temperature-health effects were identified from our systematic search using the defined MeSH terms and keywords (Fig. 1). The first screening with brief title review excluded 600 papers. Most of the excluded papers were non-relevant titles and accorded with the criteria for exclusion. The second screening, abstract review, left us with 129 papers as full-text review candidates. The most common reasons for excluding studies were: ambient temperature not used as exposure variable, use of projected and simulated temperatures, morbidity related economic loss as the outcome, comment and review papers, and non-English full text. In fact, data obtained from the abstracts of non-English papers were not sufficient for the extraction criteria described in Data Extraction. The full-text screening resulted in 64 papers remaining for inclusion in the meta-analysis. The reasons for exclusion from the study in the third screening are shown in Fig. 1.

The sixty-four studies included in the meta-analysis were summarized according to the temperature exposure in Table 1. Most of the studies in temperature-cardiovascular hospitalizations have been done in more developed and temperate countries, and the top countries that have high research intensity in temperature-cardiovascular hospitalizations are United States of America, United Kingdom, Australia, and China. Few studies have been conducted in developing and tropical

Table 1
Summaries of studies included in this review.

Authors/year	City/country	Study design/time-span	Exposure	Lags (days)	Outcome			Controlled variables
					RR/OR (95% CI)	Correspondent change in temperature/threshold	Cardiovascular morbidity	
Cold effects								
Hajat and Haines (2002)	England-London/≥65 year-old	TS/1992–1995	Mean temperature	6–15	0.978 (0.93–1.028)	1 °C decrease in temp. below 5 °C	Consultation with CVD (ICD-9, 390–403, 410–416, 420–429, 785)	Season, trend, air pollution, DOW, Christmas, New Year, Easter and bank holidays, and thunderstorm effects, consultation for influenza, humidity, interaction: temp.-humid-DOW. Humidity, air pressure
Hong et al. (2003)	Incheon/Korea	Case-crossover/1998–2000	Mean temperature (winter)	1	2.9 (1.5–5.3)	17.4 °C decreased (One interquartile)	Ischemic stroke	
Panagiotakos et al. (2004)	Athens/Greece	TS/2001–2002	Mean temperature Minimum temp.	0	1.05 (–0.91–3.01)	1 °C decrease in mean temperature	Acute coronary syndrome	DOW, season, holidays, humidity, barometric pressure, gender, age groups
Kyobutungi et al. (2005)	Heidelberg/Germany	Case-crossover/1998–2000	Mean temperature	3	2.3 (0.7–7.3)	>5 °C decrease	Ischemic Stroke	Hypertension, diabetes, smoking, previous stroke, family history of stroke, atrial fibrillation
Misailidou et al. (2006)	Multi-city/Greece	2003–2004	Mean temperature	0	1.016 (1.009–1.022)		Acute coronary syndrome	humidity
Liang et al. (2008)	Taichung/Taiwan	TS/2000–2003	Mean temperature	0	1.539 (1.16–2.1)	17 °C compared with 27–29 °C	Acute coronary syndrome	Air pollution, season, holidays
Wolf et al. (2009)	Ausburg/Germany	TS/1995–2004	Mean temperature	0	1.04 (0.97–1.12)	10 °C decrease	Myocardial Infarction	Trend, season, Monday, humidity, influenza
Lee et al. (2010)	Daegu/Korea	TS/2005–2007	Mean temperature Maximum Minimum	0 0 0	1.056 (–0.98–3.04) 1.047 (–0.91–3) 1.061 (1.04–1.09)	Decreased by 5 °C	Acute Myocardial infarction	DOW, season, holidays, humidity, wind speed, sunshine duration, thermos-hydrological index
Bhaskaran et al. (2010)	Multi-conurbation/England & Wales	TS/2003–2006	Mean temperature	0–28 2–7 8–14	1.02 (1.011–1.029) 1.006 (1.002–1.011) 1.007 (1.003–1.011)	1 °C reduction.	Myocardial infarction	Humidity, DOW, holidays, influenza, air pollution
Turner et al. (2012)	Brisbane/Australia	TS/2000–2007	Mean temperature	0–1 2–15 16–27	0.994 (0.988–1) 1.016 (1.006–1.026) 0.989 (0.979–0.998)	22 °C	CVD Ambulance attendances	Season, trend, DOW, holidays, air pollutants
Goggins et al. (2012)	Hong Kong/China	TS/1999–2006	Mean temperature	0–13 0–13	1.016 (1.0–2.2) 1.11 (1.06–1.15)	1 °C decrease from average 6.3 °C decrease from below 22	Ischemic Stroke Ischemic Stroke	Season, trend, DOW, holidays, other climate factors, and air pollutants, influenza
Wang et al. (2012)	Taiwan	TS/2000–2009	Mean temperature	0–4 0–3	1.027 (1.02–1.034) 1.07 (1.01–1.13)	1 °C decrease from average 18 °C vs. mean	Hemorrhagic Stroke ERV risk for circulatory diseases	Humidity, air pollutants
Vasconcelos et al. (2013)	Lisbon/Portugal Oporto/Portugal	TS/2003–2007	Physiologically Equivalent Temperature (PET)	0	1.022 (1.009–1.033) 1.017 (1.009–1.025)	1 °C decrease in PET during winter	Acute myocardial infarctions	DOW, holidays, season, trend, influenza, PM10
Wang et al. (2013b)	Jinan/China	TS/1990–2009	Mean temperature	0 10 0–2	1.43 (1.1–1.85) 1.08 (1.01–1.16) 1.53 (1.21–1.94)	0 °C vs. 15 °C –10 °C vs. 15 °C 0 °C vs. 15 °C	Ischemic stroke	Season, trend, DOW, holidays
Tanigawa-Sugihara et al. (2013)	Osaka/Japan	Cross-sectional analysis of prospective cohort	Mean temperature	1	1.11 (1.08–1.13) 1.16 (1.14–1.19)	5 °C decrease from 18 °C	Out-hospital Cardiac Arrest in <74 year-olds Out-hospital Cardiac Arrest in ≥74 year-olds	Humidity, pressure
Googins et al. (2013)	HongKong Taipa Kaohsiung	TS/2000–2009	Mean temperature	0–13	1.039 (1.032–1.046) 1.029 (1.021–1.037) 1.045 (1.025–1.065)	1 °C decrease from 24 °C	Acute myocardial infarctions	Season, trend, DOW, holidays, air pollutants
Son et al. (2014) Webb et al. (2014)	Multi-city/Korean Northern Territory/Australian	TS/2003–2008 Cohort/ 1992–2011	Mean temperature Minimum temp.	0–32 0	0.932 (0.857–1.014) 1.003 (1–1.006) 1.006 (0.999–1.016)	10th (2) vs. 50th (15) 5th vs. 90th 5th vs. 90th	CVD IHD Heart failure	Season, trend, humidity, DOW Non-Indigenous men Indigenous females

Mostofsky et al. (2014)	Boston/USA	Case-crossover/ 1999–2008	Mean apparent temp.	2	1.006 (1–1.012) 1.09 (1.01–1.18)	5th vs. 90th 5 °C decrease from the average	Heart failure Ischemic stroke	Non-indigenous females Humidity, Air pollution
Giang et al. (2014)	Thai Nguyen/Vietnam	TS/2008–2012	Mean temperature	0–30	1.12 (1.01–1.25)	26 °C	Myocardial infarction, angina pectoris, congestive heart failure, hypertension, stroke.	Trend, DOW, holidays
Wang and Lin (2014)	Taipei/Taiwan	TS/2000–2009	Mean temperature	0–3	1.56 (1.23–1.97) 1.78 (1.37–2.34)	14 vs. 26	Cerebrovascular diseases Hypertensive diseases	Humidity, season, trend, wind speed, DOW, holidays, influenza
Lee et al. (2014)	Korea	TS/2006–2010	Minimum temp.	5	1.01 (1–1.02)	– 1.5	Acute Myocardial infarction	Season, trend, other meteorological factors, air pollution
Urban et al. (2014)	Prague/Czech	Comparison	Mean temperature	0	1.09 (1.003–1.19)	10% coldest days	Phlebitis, thrombophlebitis	
Gomes et al. (2015)	Mabuto/Mozambique	Case-crossover/ 2005–2006	Minimum temp.	Within 7 days	1.39 (1.11–1.74)	Declines higher than 2.4 °C in minimum temp.	Stroke	Humidity, precipitation, age, smoking, cholesterol, diabetes
Cold spells								
Ma et al. (2011)	Shanghai/China	Cold-spells vs. Non-Cold spell	Maximum & Mean temperature	0	1.33 (1.28–1.37)	Maximum & average ≤3rd & 7 days	CV	
Shaposhnikov et al. (2014)	Russia	TS/1992–2005	Mean temperature	0–2	1.91 (1.07–3.41)	≤3rd & ≥5 days	Brain stroke	DOW, season, trend
Heat Effects								
Ye et al. (2001)	Tokyo/Japan	TS/1980–1995	Maximum temperature	0	0.986 (0.98–0.996)	1 °C increase	Hypertension	Season, trend, air pollution
Konken et al. (2003)	Denver, Colorado/USA	TS/1993–1997	Maximum temperature	0	1.175 (1.029–1.343)	1 °C increase	Acute myocardial infarction	Trends, DOW, air pollution
				1	0.875 (0.811–0.945)		Coronary atherosclerosis	
				1	1.132 (1.029–1.244)		Congestive heart failure	
				1	1.003 (0.999–1.007)		ICD-codes for CV and MI	
Schwartz et al. (2004)	Multi-city/USA	TS/1986–1994	Mean temperature	0	0.999 (0.997–1.000)			DOW, season, humidity, atmospheric pressure
Kovats et al. (2004)	England/1994–2000	TS/GLM	Mean temperature	0	1.17 (0.973–1.063)	24 °C	ICD-10 codes for CV	Autocorrelation, season, humidity, air pollution, influenza count
Ebi et al. (2004)	USA	TS/1983–1998	Maximum temperature	7	1.008 (1–1.017) 0.979 (0.966–0.992) 0.966 (0.949–0.984) 1.009 (1.005–1.012) 0.988 (0.986–0.990) 0.974 (0.958–0.991) 1.004 (0.999–1.010) 0.965 (0.956–0.974) 0.978 (0.965–0.992) 1.009 (1.007–1.011) 0.975 (0.972–0.977) 0.968 (0.948–0.988) 1.015 (0.994–1.038) 0.993 (0.987–0.998) 0.953 (0.922–0.984) 1.001 (1–1.002) 0.988 (0.988–0.989) 0.955 (0.943–0.967) 0.999 (0.995–1.002) 0.962 (0.960–0.964) 0.967 (0.947–0.987) 1.002 (0.999–1.004) 0.970 (0.964–0.978) 0.970 (0.964–0.978) 0.982 (0.970–0.994)	1 °C increase	Stroke	Season, trend

(continued on next page)

Table 1 (continued)

Authors/year	City/country	Study design/time-span	Exposure	Lags (days)	Outcome			Controlled variables
					RR/OR (95% CI)	Correspondent change in temperature/threshold	Cardiovascular morbidity	
Kyobutungi et al. (2005)	Heidelberg/Germany	Case-crossover	Mean temperature	0	1.04 (0.98–1.09) 2.0 (0.7–5.9)	Increase 1 °C Increase 5 °C	Stroke	Hypertension, diabetes, smoking, previous stroke, family history of stroke, atrial fibrillation
Barnett et al. (2005)	24 countries	TS/1980–1995	Mean temperature	0–3	0.992 (0.988–0.996)	1 °C increase from Mean temperature	Coronary events	Humidity
Misailidou et al. (2006)	Greece	2003–2004	Mean temperature	0	0.984 (0.978–0.991)	Change in 1 °C	Acute coronary syndrome	Humidity, region, DOW, interaction
Ren et al. (2006)	Brisbane/Australia	TS/1996–2001	Minimum temperature	0 0 1 1 2 2 0 0 1 1 1 2	0.995 (0.979–1.010) 0.996 (0.949–0.983) 1.001 (0.985–1.014) 0.992 (0.974–1.011) 1.015 (1.001–1.027) 0.986 (0.968–1.005) 0.996 (0.974–1.015) 1.005 (0.982–1.029) 1.012 (0.992–1.028) 0.992 (0.968–1.015) 1.012 (0.993–1.029) 0.979 (0.956–1.003)	19.3 °C	CVD hospital admissions	Season, trend, DOW, year, rain, humid, ozone, flu, PM
Dawson et al. (2008)	Scotland	TS/1990–2005	Maximum temperature	0 0 0	0.995 (0.987–1.003) 0.979 (0.957–1.002) 1.011 (1.001–1.022)	1 °C increase from average	IS HS Lacunar and other ischemic stroke	Season, year, DOW, pressure
			Min temperature	0 0	1.001 (0.993–1.008) 0.993 (0.974–1.014) 1.007 (0.996–1.018)		IS HS Lacunar and other ischemic stroke	
			Mean temperature	0 0 0	0.997 (0.989–1.006) 0.983 (0.96–1.007) 1.01 (0.999–1.022)		IS HS	
			Mean temperature	2 2 1 1 2 1 1	1.014 (0.993–1.035) 1.028 (0.971–1.089) 1.021 (1.007–1.035) 0.994 (0.957–1.033) 0.979 (0.939–1.02) 1.006 (0.979–1.034) 0.994 (0.982–1.005) 0.994 (0.988–1.001)		Ischemic stroke (IS) Haemorrhagic stroke (HS) Ischemic stroke Haemorrhagic stroke Lacunar and other ischemic stroke	
Michelozi et al. (2009)	12 European cities	TS/GLM 1990–2004	Maximum apparent temperature	0	0.994 (0.982–1.005) 0.994 (0.988–1.001)	32.3	ICD-9 codes for CV	Holidays, DOW, calendar month, air pollution, other meteorological factors
Wang et al. (2009)	Brisbane/Australia	TS/1996–2005	Maximum temperature	0	1 (0.970–1.020)	21	CV (stroke)	Air pollutants, humidity
Lin et al. (2009)	USA/Temperate	TS/1991–2004	Mean temperature	0 1 2 3 4	0.997 (0.987–1.007) 1.006 (0.996–1.017) 1.006 (0.996–1.016) 1.036 (1.003–1.069) 1.007 (0.985–1.030)	29.4(27.9–31)	ICD-9 codes for CVD from admissions records	Holidays, DOW, long-term trend, air pollution, atmospheric pressure
			Mean apparent temperature	0 1 2 3 4	1.002 (0.961–1.043) 1.025 (1.006–1.044) 1.022 (1.005–1.039) 1.036 (1.019–1.053) 1.014 (0.996–1.032)	35.6(32.5–38.6)		
Green et al. (2010)	USA/Temperate	Case-crossover/1999–2005	Mean apparent temperature	0	1 (0.998–1.002) 1.003 (0.999–1.007) 0.999 (0.993–1.005)	Not reported	All CVD Ischemic heart disease Acute myocardial infarction	Season, DOW, air pollution, non-linear effects

					0.981 (0.974–0.987) 0.997 (0.977–1.017)		Hemorrhagic stroke All cerebrovascular disease	
Ostro et al. (2010)	California/USA	Case-crossover/ 1999–2005	Apparent temperature	0	1.002 (1.001–1.004) 1.003 (1–1.006) 1.001 (0.997–1.004)	Not reported	CV CV (Stroke) CV (MI) CVD	Season, trend, family income, socio-economic factors
Wichmann et al. (2011)	Copenhagen/Denmark	Case-crossover/ 2002–2006	Max Apparent temperature	0–5	0.989 (0.983–0.995)	8 °C increase (1 IQR)		air pollutants, public holidays, influenza
Pudpong and Hajat (2011)	Thailand/ Tropical	Time-series/ 2002–2006	Mean temperature	0–13	0.979 (0.867–1.105)	29 °C	ICD-10 codes for CV	Season, long-term trend, number of hospital involved, month, day of week, holiday, influenza, air pollution, humidity, rainfall.
Alessandrini et al. (2011)	Emilia-Romagna/ Italy	TS/2002–2006	Mean apparent	0	1.079 (1.038–1.12)	30	Emergency ambulance dispatches	Air pollution, season, trend, holidays & weekend
Wilker et al. (2012)	Boston/USA	Repeated measure analysis	Mean apparent temperature	0–3 0–4 0–4	1.113 (1.11–1.225) 1.114 (1.012–1.225) 1.216 (1.025–1.442)	5 °C increase	Biomarkers of Heart failure	Humidity, pressure, Ozone, PM2.5
Bhaskaran et al. (2012)	Multi-conurbation/- England & Wales	Case-crossover	Mean temperature	1–6 h 7–12 h 13–18 h 19–24 h 25–48 h 49–192 h 193–360 h	1.019(1.005–1.033) 1.002 (0.991–1.014) 1.011 (0.997–1.026) 0.989 (0.997–1.001) 0.991 (0.981–1.001) 0.996 (0.986–1.006) 0.991 (0.981–1.002)	1 °C increase above 20 °C	Acute myocardial infarction	Air pollution
Basu et al. (2012)	California/USA	Case-crossover	Apparent temperature	0	1.017 (1–1.033) 1.028 (1.009–1.047) 1.028 (1.009–1.049) 1.127 (1.083–1.174) 0.9 (0.87–0.94) 0.916 (0.851–0.986) 0.864 (0.775–0.962)	10 °F increase	Ischemic heart disease Ischemic stroke Cardiac dysrhythmia Hypotension Hypertension Hemorrhagic stroke Aneurysm	Air pollution
Turner et al. (2012a)	Brisbane/Australia	TS/2000–2007	Mean temperature	0–1 2–15 16–27	1.005 (0.997–1.012) 0.982 (0.969–0.994) – 0.35 (– 1.51,0.81)	1 °C increased above 22	CVD Ambulant attendances	Season, trend, DOW, holidays, air pollution
Hori et al. (2012)	?/Japan	TS/	Mean temperature	0	1.078 (1.021–1.133) 1.36 (1.16–1.59) 1.117 (1.004–1.199)		Acute coronary syndrome Intracerebral haemorrhage Cerebral infarction	Season, tend, DOW, holidays, influenza
Williams et al. (2012)	Perth/Australia	TS/1994–2008	Max temp. Min temp.	0 0	1.022 (0.991–1.054) 1.014 (0.990–1.039)	10 °C increase	CVD ED	Air pollution
Monteiro et al. (2013)	Porto/Portugal	TS/2002–2007	Apparent temperature	0	0.975 (0.957–0.993)		Circulatory morbidity	Ozone, PM
Radisauskas et al. (2013)	Kaunas/Lithuania	TS/2000–2007	Mean temperature	0	0.97 (0.96–0.98)	5 °C increase	Acute Myocardial infarction	Atmospheric pressure
Wang et al. (2013a, 2013b)	China	TS/1990–2009	Mean temperature	0	0.43 (0.31–0.59)	30 °C vs. 15 °C	Ischemic stroke	Season, trend, DOW, holidays
Tanigawa-Sugihara et al. (2013)	Japan	Cross-sectional analysis of prospective cohort	Mean temperature	1	1.016 (0.978–1.055) 0.912 (0.876–0.950)	5 °C increase from 18 °C	Out-hospital Cardiac Arrest in <74 year-olds Out-hospital Cardiac Arrest in ≥74 year-olds	Humidity, pressure
Son et al. (2014)	Multi-city/Korea	TS/2003–2008	Mean temperature	0	4.5% (0.7–8.5)	99th (25) vs. 90th (15)	CVD	Season, trend, humidity, DOW
Giang et al. (2014)	Hanoi/Vietnam	Time-series/2008–2012	Mean temperature	0–30	1.17 (0.9–1.52)	26 °C	Myocardial infarction, angina pectoris, congestive heart failure, hypertension, stroke. Acute heart failure	Trend, DOW, holidays
Das et al. (2014)	Multi-countries	Case-crossover/ 2007–2010	Maximum temperature Mean temperature	1–3 1–3	1.18 (1.06–1.30) 1.21 (1.1–1.32)			
Ravljien et al. (2014)	Slovenia	TS/2008–2011	Mean temperature	0	0.993 (0.988–0.998)		Acute coronary syndrome	Humidity, season, atmospheric pressure

(continued on next page)

Table 1 (continued)

Authors/year	City/country	Study design/time-span	Exposure	Lags (days)	Outcome			Controlled variables
					RR/OR (95% CI)	Correspondent change in temperature/threshold	Cardiovascular morbidity	
Sheridan and Lin (2014)	New York/USA	TS/1991–2004	Hot days	0	1.002 (0.989, 1.014)		ICD-10 for CVD	
Wang and Lin (2014)	Taipei/Taiwan	TS/2000–2009	Mean temperature	0–3	2.36 (1.33–4.19)	32 vs. 26	Cerebrovascular diseases	Humidity, season, trend, wind speed, DOW, holidays, influenza
Gronlund et al. (2014)	Multi-city/USA	Case-crossover	Mean apparent temperature	0–1	0.996 (0.994–0.998)	90th vs. 75th	ICD-10 codes for CVD	
				0–7	0.987 (0.984–0.99)			
				0–1	0.984 (0.973–0.994)	99th vs. 75th		
				0–3	0.98 (0.971–0.99)			
				0–5	0.98 (0.97–0.99)			
Lee et al. (2014)	Korea	TS/2006–2010	Maximum temperature	0–7	0.982 (0.973–0.992)		Acute Myocardial infarction	Season, trend, other meteorological factors, air pollution
			Mean temperature	4	1.07 (1.05–1.10)	31.5		
Urban et al. (2014)	Prague/Czech	TS/1994–2009	Mean temperature	0	1.064 (1.004–1.128)	10% warmest days	Atherosclerosis	Season, trend, weekends, holidays
	Moscow/Russia	TS/1992–2005	Mean temperature	0–1	1.16 (1.02–1.3)	10 °C increase	Brain stroke	
Webb et al. (2014)	Australian	Cohort/1992–2011	Maximum temperature	0	1.32 (1.1–1.56)	95th vs. 90th	IDH	Indigenous females
Heatwave effects								
Semenza et al. (1999)	Chicago/USA	Comparison/heatwave vs. non-heatwave	Mean temperature	0	1.23 (1.07–1.38)	Specific time	ICD-9 codes of CVD	
Mastrangelo et al. (2007)	Veneto/Italy	Comparison/GEE model	Mean temperature	0	1.0 (0.99–1.01)	Specific heatwave periods	Circulatory	DOW
Empana et al. (2009)	Paris/France	Heatwave vs. non-heatwave	Mean temperature	0	2.34 (1.6–3.41)	Specific time (>38.1 °C)	Out-of-hospital cardiac arrest	Age, gender
Knowlton et al. (2009)	California/USA	Heatwave vs. non-heatwave	Specific time defined by the meteorologist	0	1.02 (1.01–1.03)		ED visits for CVD	State wide
					1.01 (1–1.02)		Hospitalizations	
					1.02 (0.96–1.07)		Acute MI, ED visits	
					1.02 (0.97–1.06)		Acute MI, hospitalizations	
					1.05 (1.02–1.09)		CVD, ED visits	
Wang et al. (2012)	Taiwan	TS/2000–2009	Mean temperature	0	1.23 (0.98–1.54)	99th & >3 days	Circulatory diseases	Central Coast region
Williams et al. (2012)	Perth/Australia	Heatwave vs. non-heatwave	Maximum temperature	0	1.017 (0.953–1.086)	≥35 °C & ≥3 days	CVD EDs	Air pollution
Vaneckova and Bambrick (2013)	Sydney/Australia	Case-crossover	Mean temperature	0	1.01 (1–1.02)	95th & 2–3 days	CVD	Ozone, humidity, PM
Turner et al. (2013)	Brisbane/Australia	TS/2000–2007	Maximum temperature	0	1 (0.98–1.03)	99th & 2–3 days		
Sheridan and Lin (2014)	New York/USA	TS/1991–2004	Apparent temperature	0	1.295 (1.004–1.67)	Max. temp > 37 °C for ≥2 days	Ambulance attendances	Other climates & air pollution
Ma et al. (2011)	China/Shanghai	Heatwave vs. non-heatwave	Maximum temperature	0	0.981 (0.959–1.002)	Specific time	ICD-10 for CVD	
Bobb et al. (2014)	USA/temperate	TS/1999–2010	≥99th & ≥2 days Apparent temperature	0	1.08 (1.05–1.11)	Maximum temp. >35 & mean temp >97th & ≥7 days.	ICD-10 codes for CVD	Season, trend
					0.979 (0.970–0.987)	99th & ≥2 days	CVD	
					0.996 (0.987–1.005)	Temp ≥95th & 2 days	ICD-10 codes for CVD	
					1.006 (0.994–1.018)	≥95th & 4 days		
					0.994 (0.976–1.012)	≥95th & 6 days		
Son et al. (2014)	Korea	TS/2003–2008	Mean temperature	0	0.991 (0.965–1.018)	≥95th & 8 days		Season, trend, humidity, DOW
					0.957 (0.856–1.07)	98th & ≥2 days	CVD	
Shaposhnikov et al. (2014)	Moscow/Russia	TS/1992–2005	Mean temperature	1	0.68 (0.48–0.95)	≥97th & ≥5 days	MI	DOW, season, trend
Ma et al. (2011)	China/Shanghai	Compare incidence/2005–2008	Maximum & mean temperature	0	1.33 (1.28–1.37)	≤3rd & ≥7 days	ICD-10 codes for CVD	
Phung et al, 2015	Ho Chi Minh City/Vietnam	TS/ 2004–2013	Mean temperature	0	1.129 (0.972–1.311)	≥99th & ≥2 days	ICD-10 codes for CVD	Season, trend, humidity, DOW

Diurnal temperature								
Lee et al. (2010)	Korea	TS/2005–2007	Mean temperature	0	1.068 (1.005–1.13)	Increased by 5 °C	Acute myocardial infarction	DOW, season, holidays, other climate factors
Lim et al. (2012)	Korea	TS/2003–2006	DTR	0	1.03 (1.014–1.046)	1 °C increase in DRT	Cardiac failure	Season, trend, weather, air pollution
Wang et al. (2013a)	China	TS/2009–2011	Mean temperature	0	1.004 (0.999–1.046)		CVD	
				1	1.004 (0.99–1.046)			
				2	1.002 (0.998–1.007)			
				3	1.0007 (0.997–1.005)			
				4	1.0001 (0.996–1.004)			
				5	0.997 (0.993–1.001)			
				6	0.997 (0.993–1)			
				7	1.001 (0.972–1.005)			
				0–1	1.006 (1–1.012)			
				0–2	1.008 (1.001–1.015)			
				0–3	1.007 (1–1.015)			
				0–4	1.007 (0.999–1.015)			
				0–5	1.004 (0.996–1.013)			
				0–6	1.001 (0.993–1.01)			
				0–7	1.002 (0.993–1.01)			
Shaposhnikov et al. (2014)	Russia	TS/1992–2005	Mean temperature	0–5	1.26 (1.02–1.57)	10 °C increase in DRT	Brain stroke	DOW, season, trend
Qiu et al. (2013)	China	TS/2000–2007	DTR	0	1.009 (1.003–1.014)	1 °C increase	Heart failure	Season, trend, mean temperature, humidity, air pollution
				1	1.009 (1.003–1.014)			
				2	1.008 (1.003–1.013)			
				3	1.009 (1.004–1.014)			
				4	1.006 (1.001–1.011)			
				5	1.006 (1.003–1.011)			
				0–5	1.038 (1.034–1.042)	Cumulative effect DTR > 9.6 compared with DTR < 5.8	Emergency room admissions for ACS	Season, holidays, air pollution
Liang et al. (2008)	Taiwan	TS/2000–2003	Mean temperature	0	1.34 (1–1.8)			

TS: time-series, DTR: diurnal temperature; DOW: day of week; CVD: cardiovascular diseases; RR: relative risk; OR: odds ratio.

countries (Fig. 2). Of the included studies, 11 reported the effects of cold only (Bhaskaran et al., 2010; Goggins et al., 2012; Goggins et al., 2012; Gomes et al., 2015; Hajat and Haines, 2002; Hong et al., 2003; Mostofsky et al., 2014; Panagiotakos et al., 2004; Vasconcelos et al., 2013; Wang et al., 2012; Wolf et al., 2009). Among them, only two studies reported the effects of cold spells. Therefore, cold spells were not involved in further analysis. Twenty four studies reported the effects of heat only (Alessandrini et al., 2011; Barnett et al., 2005; Basu et al., 2012; Bhaskaran et al., 2012; Das et al., 2014; Dawson et al., 2008; Ebi et al., 2004; Green et al., 2010; Hori et al., 2012; Konken et al., 2003; Kovats et al., 2004; Lin et al., 2009; Michelozzi et al., 2009; Monteiro et al., 2013; Ostro et al., 2010; Pudpong and Hajat, 2011; Radisauskas et al., 2013; Ravljen et al., 2014; Ren et al., 2006; Schwartz et al., 2004; Wang et al., 2009; Wichmann et al., 2011; Wilker et al., 2012; Ye et al., 2001), and nine studies provided evidence on heatwave-cardiovascular relationship (Bobb et al., 2014; Empana et al., 2009; Knowlton et al., 2009; Ma et al., 2011; Mastrangelo et al., 2007; Semenza et al., 1999; Turner et al., 2013; Vaneckova and Bambrick, 2013; Wang et al., 2012) while two studies reported the effects of variation in diurnal temperature only (Lim et al., 2012; Qiu et al., 2013). Seventeen studies reported the effects of multiple temperature exposure, of which nine studies examined the effects of both cold and heat exposure (Giang et al., 2014; Kyobutungi et al., 2005; Lee et al., 2014; Misailidou et al., 2006; Tanigawa-Sugihara et al., 2013; Turner et al., 2012a; Urban et al., 2014; Wang and Lin, 2014; Webb et al., 2014), two studies examined cold and diurnal temperature exposure (Lee et al., 2010; Liang et al., 2008), one study examined cold, heat and diurnal temperature exposure (Wang et al., 2013a), one study examined cold, heat and heatwave exposure (Webb et al., 2014), 3 studies examined heat and heatwave exposure (Gronlund et al., 2014; Sheridan and Lin, 2014; Williams et al., 2012), and only one study examined heat, heatwave and diurnal temperature (Shaposhnikov et al., 2014). The most commonly used temperature measurements were daily mean and maximum temperature, although some studies used minimum, apparent, and physiologically equivalent temperature (PET).

Some explicit threshold values were provided from the included studies. The mean temperature associated with cold effect ranged from 5 to 26 °C while that associated with heat effect ranged from 24 to 35.6 °C. In absence of a derived threshold, the remaining studies used average value or a specific percentile of temperature to test for the presence of temperature effect. The cold spell and heatwave definitions varied in the included studies; however the most common definitions the investigator used for these events were the long lasting extremes of temperature (e.g. ≤ 5 th or ≥ 95 th percentile for ≥ 2 days). The lag effects that were investigated in the included studies ranged from 0 to 30 days. One study, exceptionally, examined the lag effect by hours (Bhaskaran et al., 2012). Among the 64 included studies, 30 examined effect estimates for general cardiovascular diseases, and the rest examined effects for cause-specific cardiovascular diseases, including acute coronary syndrome, ischemic stroke, myocardial infarction, heart failure, hypertension, cerebrovascular disease, cardiac arrest. One study provided results in biomarkers of heart failure (Wilker et al., 2012). The most common method used to examine the relationship between temperature and cardiovascular hospitalizations was time-series using either generalized linear models (GLM) or generalized additive models (GAM) (e.g. Alessandrini et al., 2011; Barnett et al., 2005; Bhaskaran et al., 2010; Bobb et al., 2014; Dawson et al., 2008) while some studies used case-crossover design (e.g. Basu et al., 2012; Bhaskaran et al., 2012; Wichmann et al., 2011). Comparison of groups (e.g. Empana et al., 2009; Urban et al., 2014) and cohort studies (Tanigawa-Sugihara et al., 2013; Webb et al., 2014) were also employed. The most common confounding factors considered in exposure-outcome analyses comprised season and long-term trends using spline functions, days of the week, holidays, and air pollutants (Table 1). The reported OR/RR of incident cardiovascular hospitalizations in individual studies ranged from 0.93–2.9 (cold effect), 0.43–

2.36 (heat effect), 0.68–2.34 (heatwave), and 0.99–1.34 (diurnal temperature). However, the means of OR/RR are all above unity (1.19, 1.03, 1.1, and 1.03, respectively). According to our quality criteria described in Table S2, the quality score of the included papers ranged from 15 to 26, corresponding to the range of 55–96% quality criteria met by papers. This result indicated that all papers included in the meta-analysis satisfied the quality criteria (score > the cut-off points of 13) and thus possess high research value.

3.2. Temperature and risk of cardiovascular hospitalizations

The pooled effect sizes of relationship between temperature exposure and risk of cardiovascular hospitalizations are separately reported for each of the temperature exposures: cold exposure, heat exposure, heatwave, and diurnal temperature. The pooled effect size for the relationship between the change in temperature condition and the change in risk of cardiovascular hospitalizations was a 2.8% increase (RR, 1.028; 95% CI, 1.021–1.035) for cold exposure (Fig. 3), 2.2% increase (RR, 1.022; 95% CI, 1.006–1.039) for heatwave exposure (Fig. 4), and 0.7% increase (RR, 1.007; 95% CI, 1.002–1.012) for an increase in diurnal temperature (Fig. 6). In contrast, no effect of heat exposure on the risk of cardiovascular hospitalizations was observed when all studies were included (RR, 0.997; 95% CI, 0.994–0.999) (Fig. 5). However, when incorporating the variation of effect sizes from different settings using the EB approach, the pooled effect sizes were increased for all types of temperature exposure (Fig. 7), comprising: 7.8% (RR, 1.078; 95% CI, 1.074–1.081) for cold exposure, 1% (RR, 1.01; 95% CI, 1.008–1.011) for heat exposure, 6.1% (RR, 1.061; 95% CI, 1.053–1.07) for heatwave exposure, and 1.5% (RR, 1.015; 95% CI, 1.011–1.02) for an increase in diurnal temperature. The significant heterogeneity between studies was found with all of the temperature exposure categories. The percentages of total variation across these studies caused by heterogeneity were high: I^2 , 92.2% for cold exposure, 95.7% for heat exposure, 92.6% for heatwave, and 92.5% for diurnal temperature. These reflected inconsistent results of ORs/RRs among the included studies. The Begg's funnel plot with pseudo 95% confidence limits and Egger's test revealed potential publication bias for cold exposure (Egger's bias, 2.5, $p < 0.05$) while no evidence of publication bias was found for the heat exposure, heatwave, or diurnal temperature (Egger's bias, 2.5, $p = 0.4$; Egger's bias, 0.43, $p = 0.4$; Egger's bias, 1.95, $p = 0.9$; Egger's bias, 1.08, $p = 0.5$, respectively).

The dose-response effects of temperature, lag and latitude on the risk of cardiovascular hospitalizations are shown in Table 2. The results showed that the risk of cardiovascular hospitalizations significantly increase with each degree (1 °C) decrease in temperature in cold exposure (0.8%, 95% CI: 0.08–1.5) and with each degree (1 °C) increase in diurnal temperature (1.9%, 95% CI: 0.5–3.3), whereas the risk of cardiovascular hospitalizations decreased with each degree (1 °C) increase in temperature in heat exposure (–0.5, 95% CI: –0.9–0.04). However, this dose-response relationship was not statistically significant. An increase in one-day lag was associated with a borderline statistically significant (at 92% & 93%) reduction in risk of cardiovascular hospitalizations for cold exposure (–0.6%, 95% CI: –1.2–0.08) and diurnal temperature (–0.2%, 95% CI: –0.4–0.01) while no significant effect was seen in heat exposure. The increase of one-degree in latitude was found to statistically associate with a decrease in risk of cardiovascular hospitalizations (–0.05, 95% CI: –0.09–0.01) for diurnal temperature only. For all temperature exposures, I^2 values were mostly on the order of 88% to 100%, indicating large between-study heterogeneity, and supporting the use of random effect models.

Sensitivity analysis with the exclusion of studies with quality score of less than 20 points shows a very subtle change in the effects of cold exposure (RR, 1.029 vs. 1.028) and no difference for heat, heatwave, and diurnal temperature exposure. Subgroup analysis, which was performed for lag 0-day only, provided a higher risk of cardiovascular hospitalizations for heatwave exposure (RR, 1.024, 95% CI: 1.008–1.040)

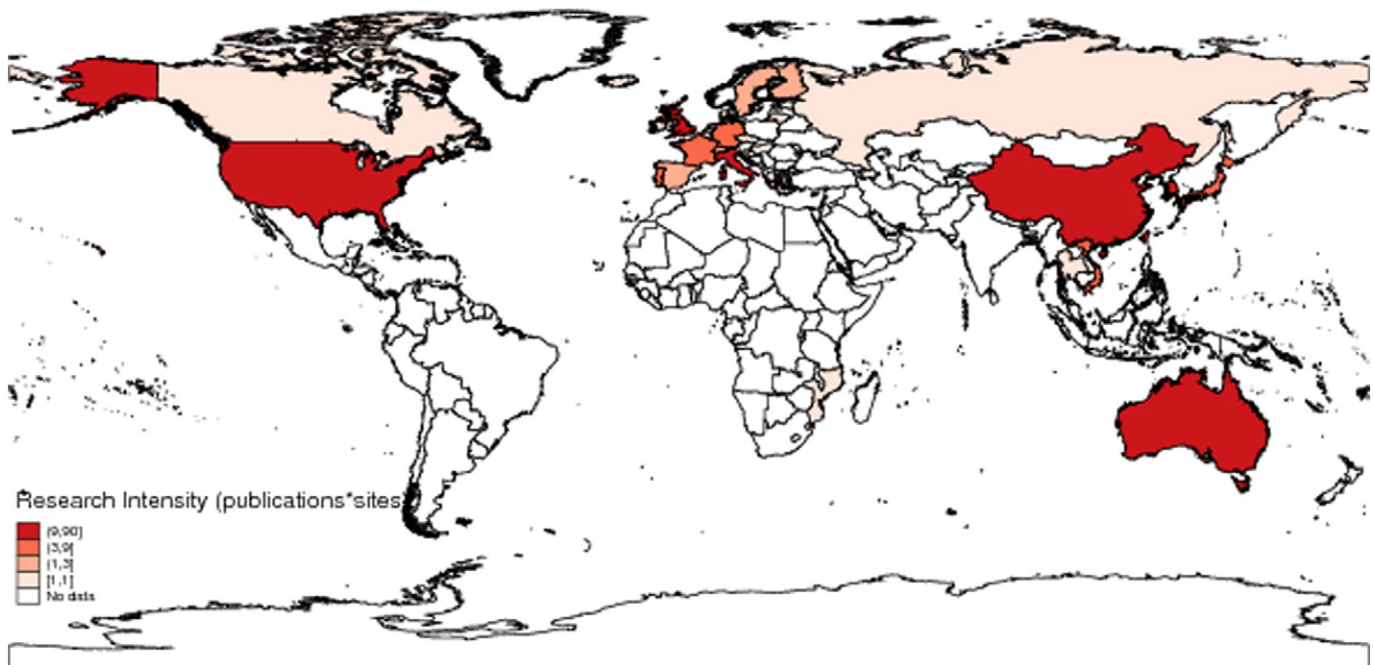


Fig. 2. Research Intensity (number of publications x number of sites) on the relationship between ambient temperature and cardiovascular morbidity in the world.

and diurnal temperature (RR, 1.019, 95% CI: 1.001–1.036) in comparison with the overall pooled effect sizes, whereas the lag-0 effect was lower for cold exposure (RR, 1.015, 95% CI: 1.008–1.023). The sensitivity analysis with the exclusion of the studies using CVD consultation and CVD-biomarkers as the outcomes provided the same results as those obtained for overall pooled effect sizes (the results are not shown).

4. Discussion

This meta-analysis offers an updated quantitative examination of the effects of different ambient temperature conditions, including cold, heat, heatwave, and diurnal exposure on risk of cardiovascular hospitalizations. The strength of the study is to fill the gaps of knowledge about temperature-cardiovascular morbidity reported from previous studies (Astrom et al., 2011; Bhaskaran et al., 2009; Turner et al., 2012b; Ye et al., 2012) which did not provide the quantitative pooled effect sizes of the temperature-cardiovascular morbidity relationship for separated temperature exposure conditions. In this study, we found the significantly increased risk of cardiovascular hospitalizations in relation to cold, heatwave exposure, and variation in diurnal temperature. However, the relationship between heat exposure and cardiovascular hospitalizations was found to be inconsistent. A reduction in risk of cardiovascular hospitalizations has been found when lags are longer. Heatwave exposure causes more immediate effects than cold exposure. Latitude modifies the temperature-cardiovascular hospitalization relationship with variation in diurnal temperature only.

The significant effect of cold exposure on elevated risk of cardiovascular diseases is consistent with the findings of a majority of previous studies, in which increased risk of cardiovascular hospitalizations was associated with a reduction in temperature during the cold seasons. The risk of ischemic stroke (IS) increases by 2%–53% with each 1 °C decrease from the average and threshold temperature (Goggins et al., 2012; Mostofsky et al., 2014; Wang et al., 2013b), and the risk of IS elevated from 130% (Kyobutungi et al., 2005) to 190% (Hong et al., 2003) when the temperature reduces from 5 °C to 17 °C. A decrease of 1 °C in average temperature causes an increase of 1.6–6% in risk of acute coronary syndrome (ACS) (Misailidou et al., 2006; Panagiotakos et al., 2004), and a decrease of 10 °C causes an increase of up to 54% in risk of ACS (Liang et al., 2008). Likewise, the risk of myocardial infarction (MI) rises by 0.6–12% for

each 1–5 °C decrease from average and threshold temperature (Bhaskaran et al., 2010; Giang et al., 2014; Goggins et al., 2013; Lee et al., 2014; Vasconcelos et al., 2013; Wolf et al., 2009). The risk of emergency admissions due to cerebrovascular and hypertensive diseases increases by 56–78% when the temperature decreases 12 °C from the threshold temperature (26 °C) (Wang and Lin, 2014). The risk of cardiac arrest among the elder population was found to increase by 11–16% when the temperature went down by 5 °C from a threshold of 18 °C in Japan (Tanigawa-Sugihara et al., 2013). Several plausible mechanisms are potentially involved in increasing the risk of cardiovascular hospitalizations due to cold exposure. A potential pathological mechanism is that cold exposure causes different effects on the cardiovascular system, possibly mediated by stimulation of both sympathetic nervous activity and the coagulation system (DASH Collaborative Research Group et al., 2002; Kawahara et al., 1989; Keatinge et al., 1984; Touitou et al., 1986; Wilmschurst, 1994). The low air temperatures produce vasoconstriction leading to an increase in arterial pressure and in circulating levels of catecholamines, which can increase heart rate and cardiac work. These phenomena result in a greater oxygen demand and a potentially ischemic reaction in the vulnerable myocardium (Opie, 1998). Furthermore, cold temperatures are associated with an increase in blood pressure variability, resulting in higher cardiovascular hospitalizations and mortality (DASH Collaborative Research Group et al., 2002). In addition, the association between temperature and ACS may be indirectly causal as indoor smoking, lowliness and bodyweight increase during the winter (Wilmschurst, 1994).

In this study, we found that the heat exposure is inconsistently associated with risk of cardiovascular hospitalizations. However heatwaves, extended periods of heat exposure, are significantly associated with elevated risk of cardiovascular hospitalizations. The finding of the relationship between heat exposure and cardiovascular hospitalization is in lines with the recent review conducted by Turner et al. (2012b), in which the result of a meta-analysis indicated no apparent association between increased ambient temperature and cardiovascular morbidity (−0.5% change in risk of cardiovascular morbidity, 95%CI: −3%–2.1%) (Turner et al., 2012b). The inconsistent findings in this relationship were also reported in many original studies which suggested weak or absent association between heat exposure and cardiovascular morbidity (Barnett et al., 2005; Basu et al., 2012; Bhaskaran et al., 2012; Dawson

et al., 2008; Ebi et al., 2004; Green et al., 2010; Gronlund et al., 2014; Konken et al., 2003; Lin et al., 2009; Michelozzi et al., 2009; Ostro et al., 2010; Radisauskas et al., 2013; Ren et al., 2006; Schwartz et al., 2004; Tanigawa-Sugihara et al., 2013; Turner et al., 2012a; Wang et al., 2009; Wichmann et al., 2011; Ye et al., 2001). One possible explanation for the difference between cold- and heat-cardiovascular morbidity is that decreased cardiovascular performance is less prevalent in the warmer months of the year (Turner et al., 2012b). Moreover, the CVD diseases inconsistently reacted to an increase in temperature. For example, the study by Konken et al. (2003) illustrated that higher

temperatures are associated with increasing the frequency of hospitalization for acute myocardial infarction and congestive heart failure but did not influence visits for cardiac dysrhythmias. In contrast, higher temperature appeared to be a protective factor for coronary atherosclerosis and pulmonary heart diseases. This may be because people with chronic cardiovascular conditions avoid outdoor exposure during periods of peak heat. In addition, other factors such as age, gender and geographic region, which may modify the relationship between heat exposure and cardiovascular morbidity, were much varied among the included studies.

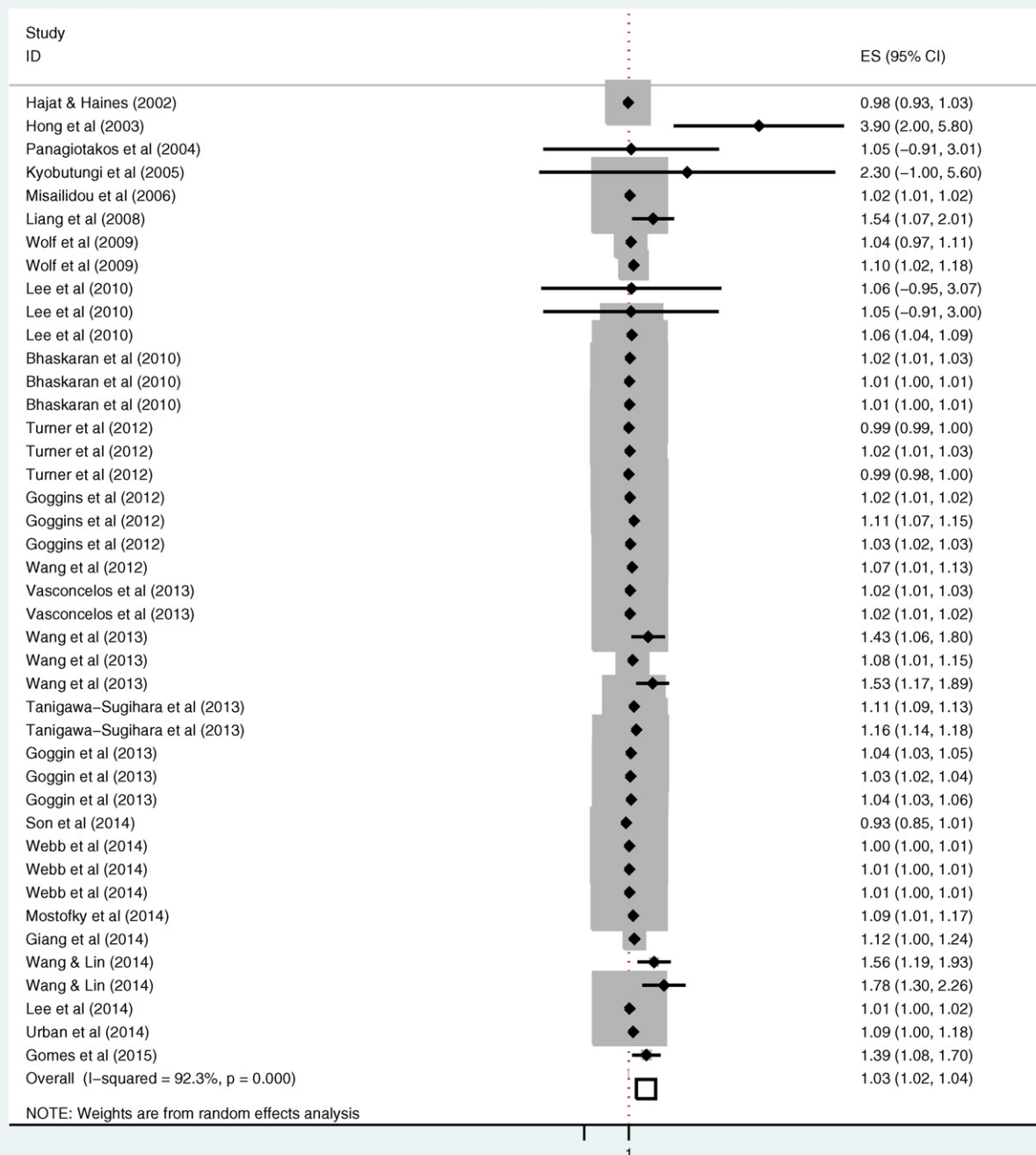


Fig. 3. Meta-analysis of cold exposure (defined as number of degrees below the defined threshold or average value, or comparison between extreme cold condition and the reference value, e.g. 5th vs. 90th) on the risk of cardiovascular hospitalization (Relative Risk & 95%CI).

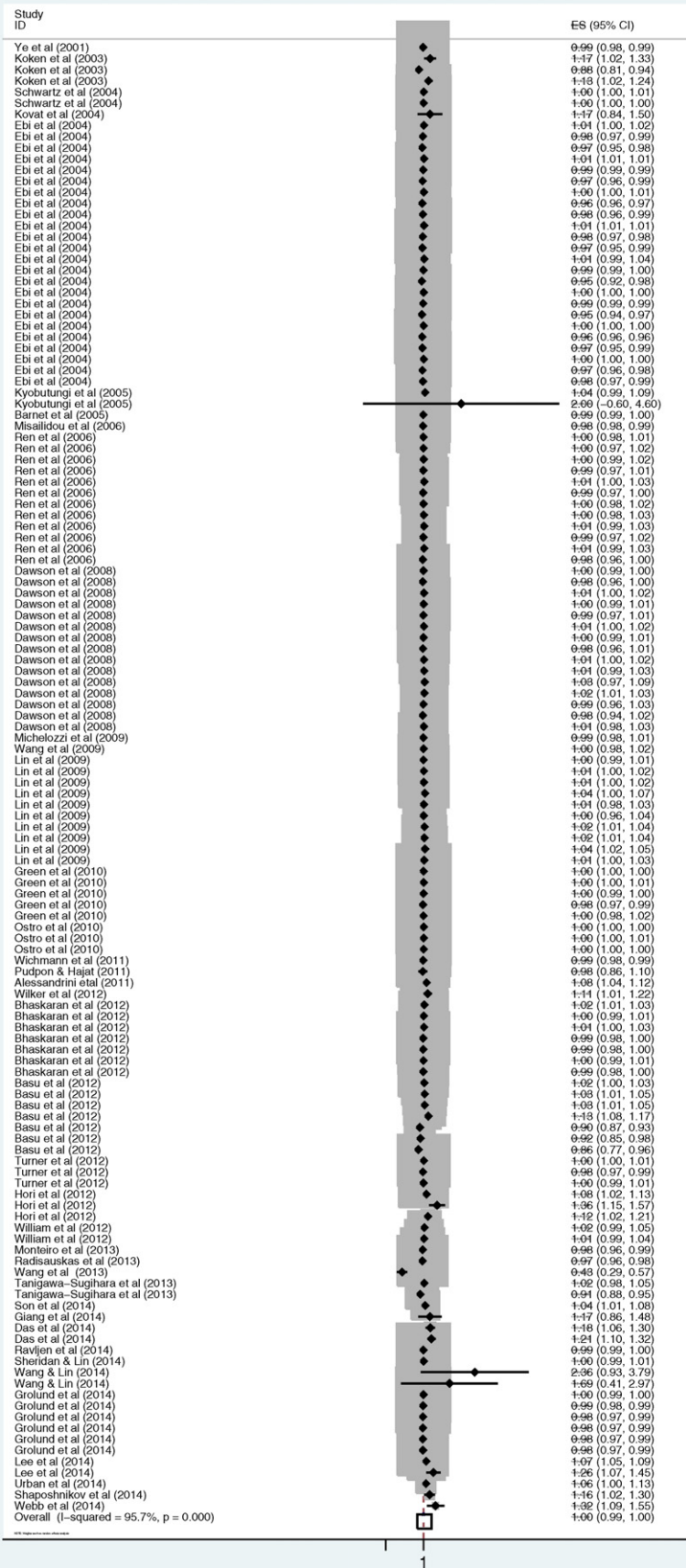


Fig. 4. Meta-analysis of heat exposure [defined as number of degrees above the defined threshold or average value but NOT during the cold season (e.g. winter), or comparison between extreme hot condition and the reference value (e.g. 99th vs. 75th)] on the risk of cardiovascular hospitalization (Relative Risk & 95%CI).

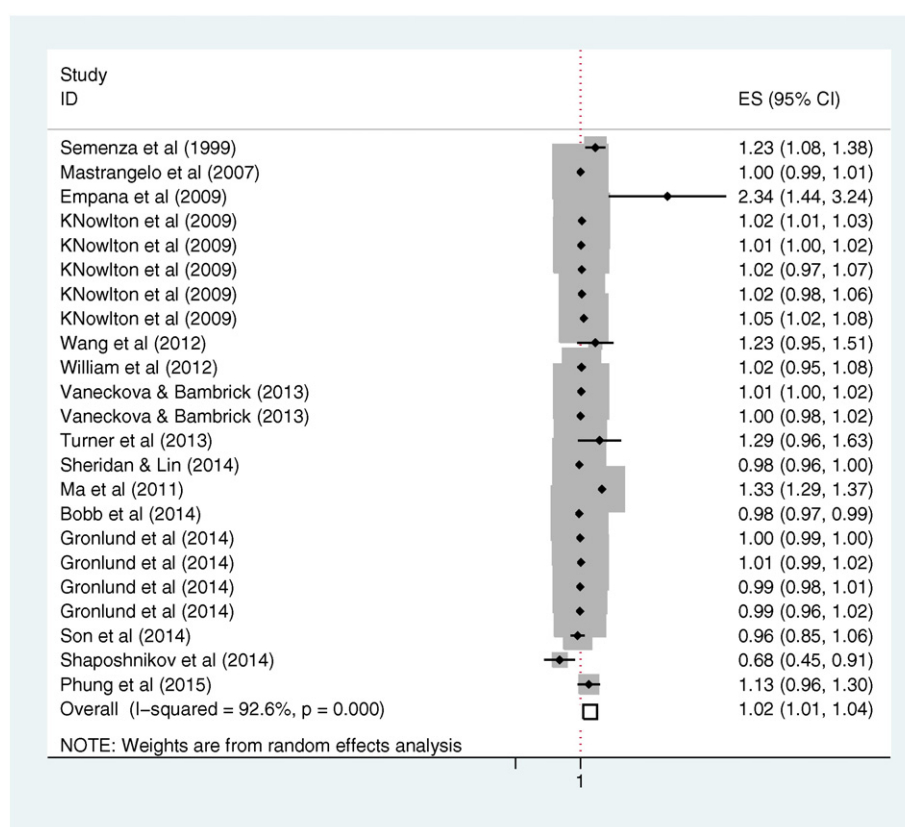


Fig. 5. Meta-analysis of heatwave exposure (defined as ≥ 2 days of extreme high temperature, e.g. 95th) on the risk cardiovascular hospitalizations (Relative Risk & 95%CI).

In contrast to heat exposure, this study found a positive effect on the risk of cardiovascular hospitalizations from heatwave. This finding is consistent with the results of previous studies which found that long lasting extremely high temperatures were associated with the elevated risk of cardiovascular hospitalizations (Empana et al., 2009; Knowlton et al., 2009; Ma et al., 2011; Semenza et al., 1999; Turner et al., 2013; Wang et al., 2012; Williams et al., 2012). However, the previous study indicated that the cardiovascular disease was not a primary cause for admission, but as a pre-existing condition it was a risk factor for hospitalization, since diagnoses of chronic cardiovascular diseases (e.g. hypertensive disease) were more common than acute cardiac events (e.g. myocardial infarction). Elderly people (>65 year-olds) were more susceptible than other age groups. This can be explained by the physiological mechanisms of chronic heart insufficiency and/or the inability to increase cutaneous circulation, which can impede dissipation of extreme heat, resulting in increased risk of hospitalization. In addition, left ventricular diastolic filling declines with age as do cardiac output and heart rate (Schulman et al., 1992; Semenza et al., 1999).

The variation in diurnal temperature was found to significantly influence the risk of cardiovascular hospitalization in this study. This was consistent with the evidence provided from the original studies in different populations and settings. The studies (Qiu et al., 2013; Wang et al., 2013a) indicated that a one degree increase in diurnal temperature causes an increase by 0.35–3.8% in risk of cardiovascular hospitalization. The risk of cardiovascular hospitalization could rise by 6.8–34% if the diurnal temperature increases more than 5 °C (Lee et al., 2010; Liang et al., 2008; Shaposhnikov et al., 2014). Previous studies also found a significant association between diurnal temperature and cardiovascular mortality (Kan et al., 2007; Tam et al., 2009). These previous studies indicated that the most vulnerable group to diurnal temperature was people aged >40 years with pre-existing cardiovascular conditions (e.g. hypertension, high serum cholesterol level) (Shinkawa et al., 1990).

The biological mechanism explaining this relationship is that a large change in temperature within one day may cause a sudden change in the heart rate and circulation of elderly people, and this phenomenon can act to increase the risk of cardiopulmonary and other diseases, even resulting in fatalities (Kenney and Hodgson, 1987).

In terms of latitude effect, this study found small negative effect of latitude on the risk of cardiovascular hospitalization for diurnal temperature exposure but not for cold and heat exposure. That means higher latitude countries show lower effects of diurnal temperature on the risk of cardiovascular hospitalization. This finding does not support the evidence from the previous review (Turner et al., 2012b) which indicated that there was an association with increased heat exposure at higher latitudes (colder climates) and risk of cardiovascular hospitalizations. The reasons for the latitude effect in colder climates were that the adaptive capacity of cold climate population is lower because the population is less acclimatized to high temperatures, live in houses that are unsuitable for hot weather, and have a lack of adaptive measures such as conditioning (Turner et al., 2012b). In terms of lag effect, this study found that the effects of exposure to hot temperatures were more immediate than for cold temperatures since the lag-0 effect of heatwave exposure was bigger than the overall effect (including all lags in analysis), whereas for cold exposure the lag-0 effects were the opposite. This finding was consistent with the results of the previous review conducted by Ye et al. (2012) which demonstrated more immediate effects of heat exposure in comparison with cold exposure. Most of the recent studies provided evidence on the short-term effects of high temperature on the same day and the 3 days following heat exposure (Green et al., 2010; Konken et al., 2003; Lin et al., 2009) while the effects of a longer lag were not clearly demonstrated from the studies (Ye et al., 2012). The more immediate effect of hot temperatures could also be an explanation for inconsistent findings in the relationship between heat exposure and cardiovascular hospitalizations, since heat exposure might cause an increase in out-of-hospital deaths before medical

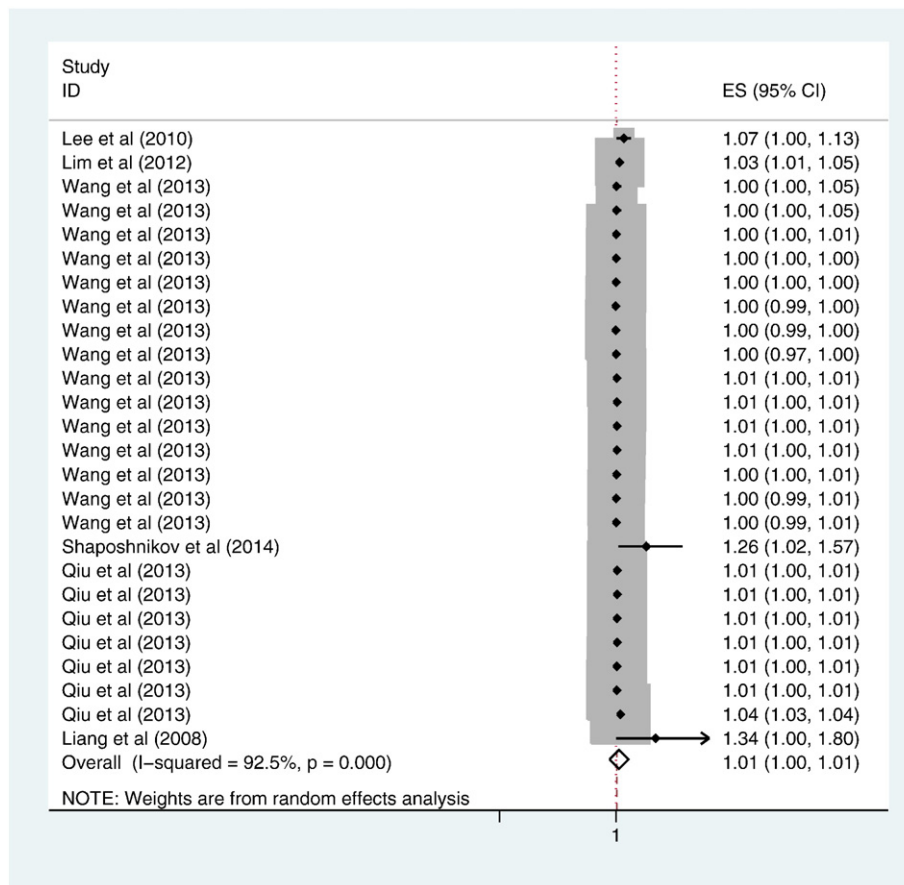


Fig. 6. Meta-analysis of diurnal temperature (defined as number of degrees between maximum and minimum temperature during the day) on the risk cardiovascular hospitalizations (Relative Risk & 95%CI).

treatment for acute cardiovascular events (Michelozzi et al., 2009; O'Neill and Ebi, 2009).

This study has some limitations. First, although an increased number of studies of the relationship between temperature and cardiovascular morbidity have been conducted recently, they were conducted predominantly in developed and temperate areas. Therefore, the pooled effect sizes could not be computed for specific geographical areas or for a limited range of climatic conditions, specifically for developing and tropical countries. Second, the variety of definitions in temperature exposure in individual included studies prevented the computation of normalized

effect estimates for all exposures in step 1. Third, the number of included studies is not sufficient to examine the cause-specific cardiovascular disease but only all-cause cardiovascular hospitalizations. The previous review (Hajat and Haines, 2002) suggested that general practitioner consultations should be considered as the outcome for better strategies in early detection of temperature-related morbidity. Nevertheless, only one study using such outcome has been conducted since 2002. Fourth, the variety of latitudes, lags, temperature measurements and thresholds used in the included studies make the pooled estimates highly heterogeneous. However, this study used a Bayesian approach to incorporate

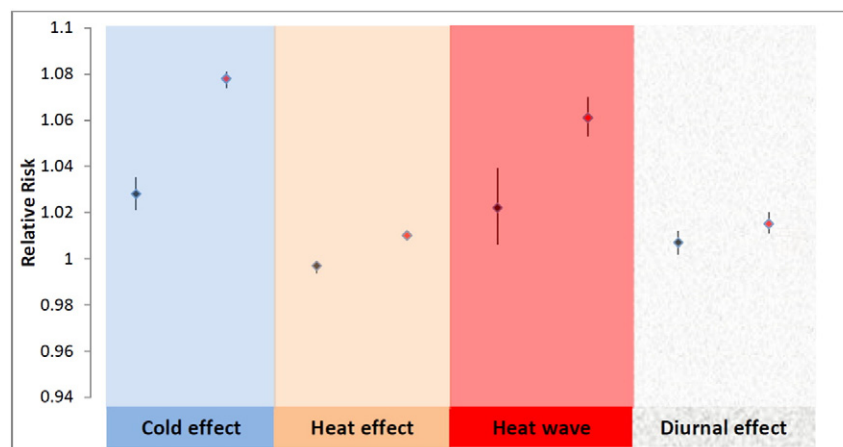


Fig. 7. Pooled effect sizes estimated by the normal and Empirical Bayesian approaches (blue: normal; red: Bayesian).

Table 2

The dose–response relationship between the risk of cardiovascular hospitalizations (%) and temperature, lag and latitude.

Temperature exposure	One-degree change in temperature ^a	95%CI	One-day increase in lag	95%CI	One-degree increase in latitude	95% CI
Cold exposure	0.8^b	0.08–1.5	−0.6^c	−1.2–0.08	−0.07	−0.2–0.1
Heat exposure	−0.5	−0.9–0.04	−0.2	−0.4–0.1	−0.005	−0.04–0.04
Diurnal temperature	1.9^b	0.5–3.3	−0.2^c	−0.4–0.01	−0.05^b	−0.09–(−0.01)

^a One-degree decrease for cold exposure, and one-degree increase for heat and diurnal exposure.^b Statistically significant at 95%.^c Statistically significant at 92%.

the variation of these factors of included studies into the meta-analysis. Fifth, there is a lack of data on socioeconomic factors and adaptive capacity which can modify the relationship between temperature and cardiovascular morbidity to an important degree. A future study should be conducted to examine temperature–cardiovascular admissions queried by more potential variables such as geographical regions, age, sex, and other socio-economic factors.

For dose–response relationship between temperature and risk of CVD admissions, one-degree change for each temperature condition (cold, heat, diurnal) may have different meaning in different climate regions, and adjusting for latitude only might not completely remove the effects of different geographical difference. Future studies are needed to better understand the dose–response relationship in different climate regions. As the findings of this study indicated that heatwaves significantly associated with elevated risk of cardiovascular admission, it would be valuable to explore and to comment on the sensitivity of the result to selected metrics (such as maximum or minimum daily temperature), and derived indexes, such as apparent temperature, as well as the lag duration. Finally, the review may not be entirely comprehensive because non-English publications elsewhere such as Central-America, Middle East were not evaluated.

5. Conclusion

This study has demonstrated the significant relationship between cold exposure, heat waves, and variation in diurnal temperature and the elevated risk of cardiovascular hospitalization. However there is an inconsistent effect of heat exposure on cardiovascular hospitalizations. Such findings have not been quantitatively estimated from the previous reviews. Based on the evidence reviewed, there is need for future studies to focus on specific geographical and climate areas, particularly in developing and tropical countries. Additionally, temperature measurements and thresholds need to be standardized for all studies in temperature and health-related effect. For example, definitions of heat waves should be homogenous for different research populations and settings. The modification effects of socio-economic factors also need to be considered more thoroughly in future studies. In terms of intervention, early warning models for temperature-related cause-specific morbidity should be developed as an effective measure for climate change adaptation strategies and heat-related morbidity prevention.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <http://dx.doi.org/10.1016/j.scitotenv.2016.01.154>.

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